EXISTENCE OF A GENERATING SEQUENCE WHOSE SUCCESSIVE QUOTIENTS ARE THE RESIDUE FIELD AT A MINIMAL PRIME OF THE SUPPORT OF A FINITE MODULE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02QC (sources/stacks-chow/pinned-original-source.tex line 17964).

Allowed standard prerequisites (editorial).

- Support of a finite module over a Noetherian local ring and finiteness of the length at a minimal prime of the support. Algebra, Lemma labelled algebra-lemma-support-point (Stacks tag 00L5), invoked at proof line 17982. Supplies that q is a minimal prime of the support exactly when M_q has finite length, which is what makes l = length_{R_q}(M_q) finite and the sequence y_1, ..., y_l available; the lifting and clearing of denominators is inside the counted span.

Source reference. Original lemma 00L5: sources/stacks-chow/tags.txt, source lines 775–775. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 17982 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let R be a local Noetherian ring. Let $\mathfrak{q} \subset R$ be a prime ideal. Let M be a finite R-module such that $\mathfrak{q}$ is one of the minimal primes of the support of M. Then there exist $x_1, \ldots, x_l \in M$ such that*

(1) *the support of $M/\langle x_1, \ldots, x_l\rangle$ does not contain $\mathfrak{q}$, and*
(2) *$\langle x_1, \ldots, x_i\rangle/\langle x_1, \ldots, x_{i-1}\rangle \cong R/\mathfrak{q}$ for $i = 1, \ldots, l$.*

Moreover, in this case $l = \mathit{length}_{R_\mathfrak{q}}(M_\mathfrak{q})$.

Proof. The condition that $\mathfrak{q}$ is a minimal prime in the support of M implies that $l = \text{length}_{R_\mathfrak{q}}(M_\mathfrak{q})$ is finite (see Algebra, Lemma 00L5). Hence we can find $y_1, \ldots, y_l \in M_\mathfrak{q}$ such that $\langle y_1, \ldots, y_i\rangle/\langle y_1, \ldots, y_{i-1}\rangle \cong \kappa(\mathfrak{q})$ for $i = 1, \ldots, l$. We can find $f_i \in R$, $f_i \notin \mathfrak{q}$ such that $f_i y_i$ is the image of some element $z_i \in M$. Moreover, as R is Noetherian we can write $\mathfrak{q} = (g_1, \ldots, g_t)$ for some $g_j \in R$. By assumption

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$g_j y_i \in \langle y_1, \ldots, y_{i-1} \rangle$ inside the module $M_{\mathfrak{q}}$. By our choice of z_i we can find some further elements $f_{ji} \in R$, $f_{ij} \notin \mathfrak{q}$ such that $f_{ij} g_j z_i \in \langle z_1, \ldots, z_{i-1} \rangle$ (equality in the module M). The lemma follows by taking

$$x_1 = f_{11} f_{12} \ldots f_{1t} z_1, \quad x_2 = f_{11} f_{12} \ldots f_{1t} f_{21} f_{22} \ldots f_{2t} z_2,$$

and so on. Namely, since all the elements f_i, f_{ij} are invertible in $R_{\mathfrak{q}}$ we still have that $R_{\mathfrak{q}} x_1 + \ldots + R_{\mathfrak{q}} x_i / R_{\mathfrak{q}} x_1 + \ldots + R_{\mathfrak{q}} x_{i-1} \cong \kappa(\mathfrak{q})$ for $i = 1, \ldots, l$. By construction, $\mathfrak{q} x_i \in \langle x_1, \ldots, x_{i-1} \rangle$. Thus $\langle x_1, \ldots, x_i \rangle / \langle x_1, \ldots, x_{i-1} \rangle$ is an R-module generated by one element, annihilated $\mathfrak{q}$ such that localizing at $\mathfrak{q}$ gives a q-dimensional vector space over $\kappa(\mathfrak{q})$. Hence it is isomorphic to $R/\mathfrak{q}$. $\square$